

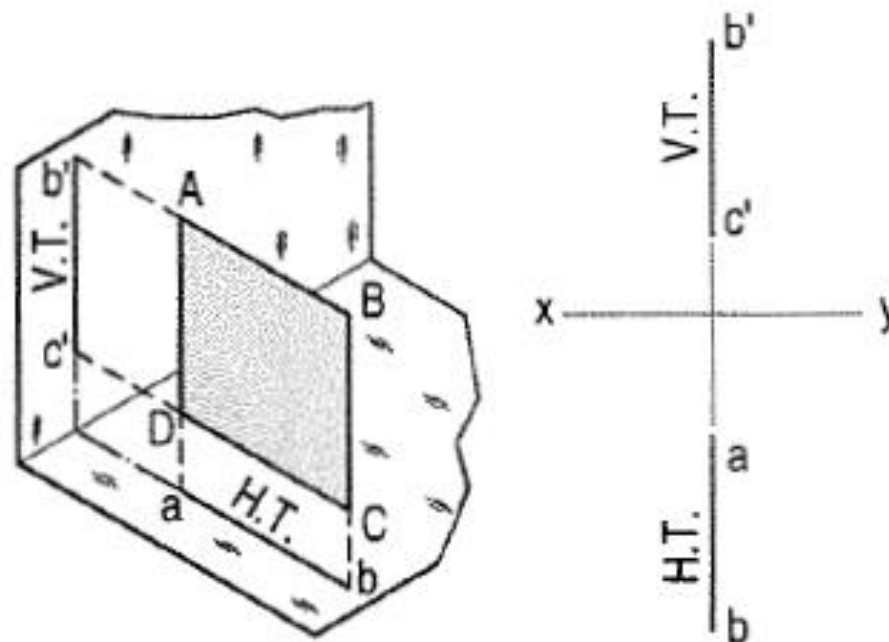
ENGINEERING DRAWING- LECTURE 2

PLANES

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Perpendicular Planes

(i) *Perpendicular to both the reference planes* (fig. 12-1): A square $ABCD$ is perpendicular to both the planes. Its H.T. and V.T. are in a straight line perpendicular to xy .



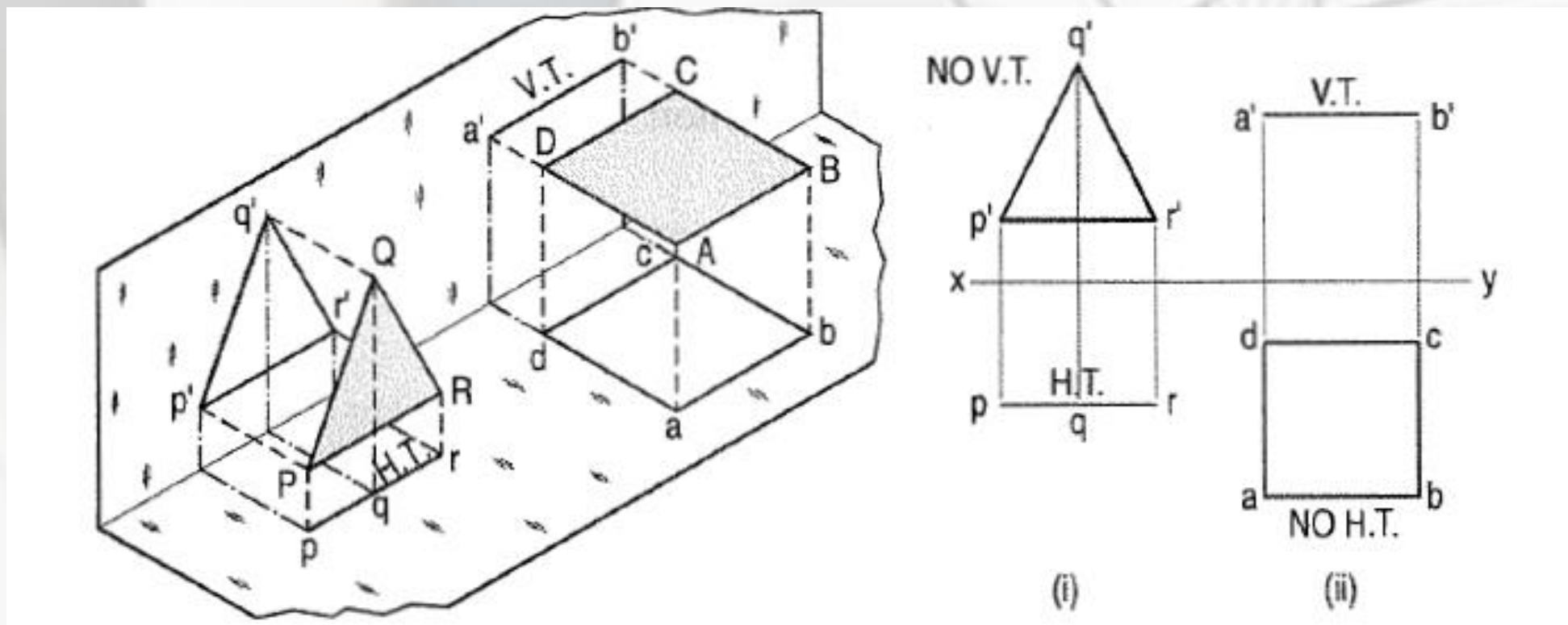
Perpendicular Planes

(ii) *Perpendicular to one plane and parallel to the other plane:*

(a) Plane, perpendicular to the H.P. and parallel to the V.P. [fig. 12-2(i)].

A triangle PQR is perpendicular to the H.P. and is parallel to the V.P. Its H.T. is parallel to xy . It has no V.T.

The front view $p'q'r'$ shows the exact shape and size of the triangle. The top view pqr is a line parallel to xy . It coincides with the H.T.



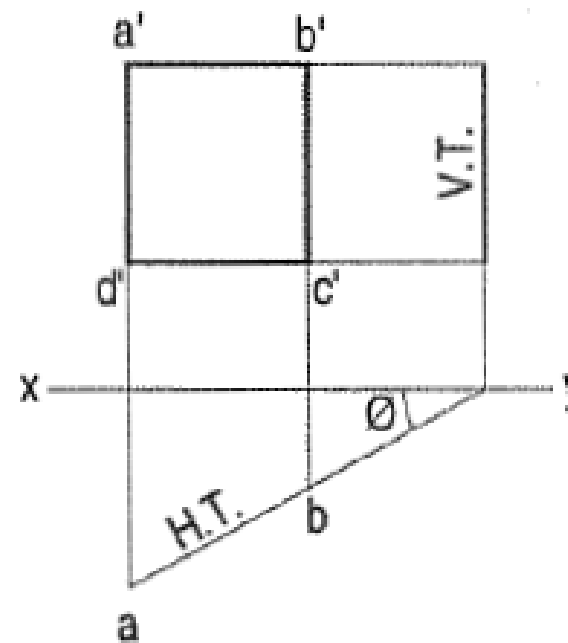
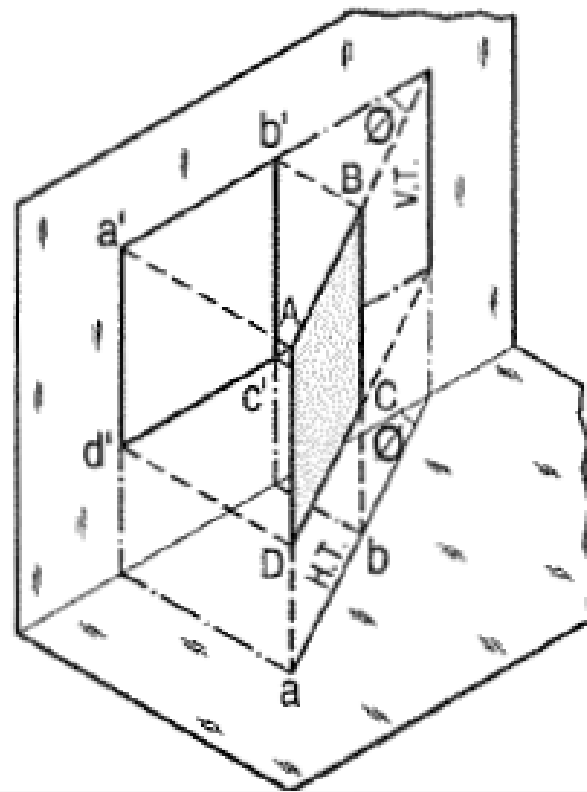
Perpendicular Planes

(iii) *Perpendicular to one plane and inclined to the other plane:*

(a) Plane, perpendicular to the H.P. and inclined to the V.P. (fig. 12-3).

A square $ABCD$ is perpendicular to the H.P. and inclined at an angle θ to the V.P. Its V.T. is perpendicular to xy . Its H.T. is inclined at θ to xy .

Its top view ab is a line inclined at θ to xy . The front view $a'b'c'd'$ is smaller than $ABCD$.



PRACTICE DRAWING 1

Problem 12-2. (fig. 12-8): An equilateral triangle of 50 mm side has its V.T. parallel to and 25 mm above xy . It has no H.T. Draw its projections when one of its sides is inclined at 45° to the V.P.

As the V.T. is parallel to xy and as there is no H.T. the triangle is parallel to the H.P. Therefore, begin with the top view.

- (i) Draw an equilateral triangle abc of 50 mm side, keeping one side, say ac , inclined at 45° to xy .
- (ii) Project the front view, parallel to and 25 mm above xy , as shown.

(2) When the plane is parallel to the V.P.: Beginning should be made with the front view and the top view projected from it.

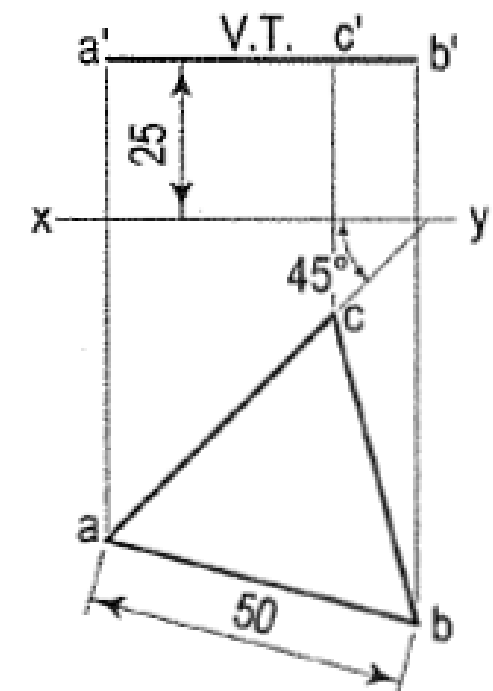


FIG. 12-8

PLANES INCLINED TO ONE AND PERPENDICULAR TO OTHER

(1) Plane, inclined to the H.P. and perpendicular to the V.P.: When the plane is inclined to the H.P. and perpendicular to the V.P., in the initial stage, it is assumed to be parallel to the H.P. Its top view will show the true shape. The front view will be a line parallel to xy . The plane is then tilted so that it is inclined to the H.P. The new front view will be inclined to xy at the true inclination. In the top view the corners will move along their respective paths (parallel to xy).

PRACTICE DRAWING -2

Problem 12-4. (fig. 12-10): A regular pentagon of 25 mm side has one side on the ground. Its plane is inclined at 45° to the H.P. and perpendicular to the V.P. Draw its projections and show its traces.

Assuming it to be parallel to the H.P.

- (i) Draw the pentagon in the top view with one side perpendicular to xy [fig. 12-10(i)]. Project the front view. It will be the line $a'c'$ contained by xy .
- (ii) Tilt the front view about the point a' , so that it makes 45° angle with xy .
- (iii) Project the new top view $ab_1c_1d_1e$ upwards from this front view and horizontally from the first top view. It will be more convenient if the front view is reproduced in the new position separately and the top view projected from it, as shown in fig. 12-10(ii). The V.T. coincides with the front view and the H.T. is perpendicular to xy , through the point of intersection between xy and the front view-produced.

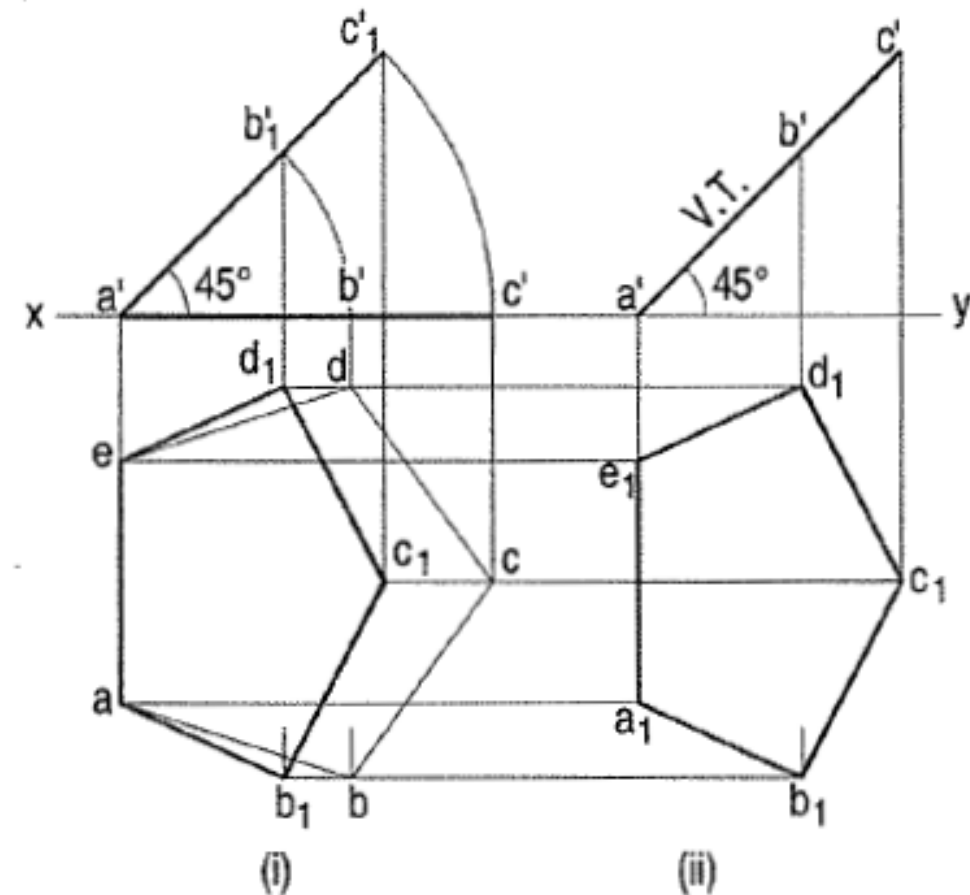


FIG. 12-10

ASSIGNMENT DRAWING -1

(2) Plane, inclined to the V.P. and perpendicular to the H.P.: In the initial stage, the plane may be assumed to be parallel to the V.P. and then tilted to the required position in the next stage. The projections are drawn as illustrated in the next problem.

Problem 12-5. (fig. 12-11): Draw the projections of a circle of 50 mm diameter, having its plane vertical and inclined at 30° to the V.P. Its centre is 30 mm above the H.P. and 20 mm in front of the V.P. Show also its traces.

A circle has no corners to project one view from another. However, a number of points, say twelve, equal distances apart, may be marked on its circumference.

- (i) Assuming the circle to be parallel to the V.P., draw its projections. The front view will be a circle [fig. 12-11(i)], having its centre 30 mm above xy . The top view will be a line, parallel to and 20 mm below xy .
- (ii) Divide the circumference into twelve equal parts (with a 30° - 60° set-square) and mark the points as shown. Project these points in the top view. The centre O will coincide with the point 4.

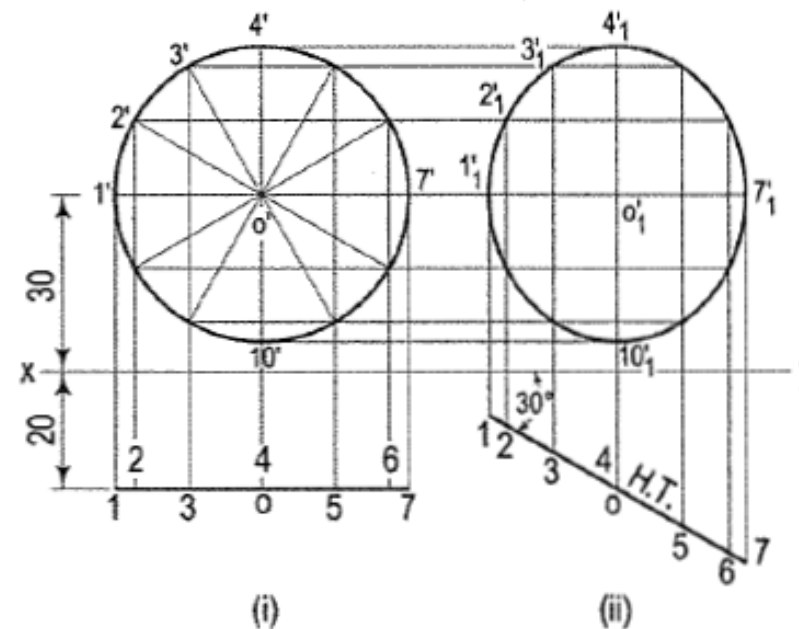


FIG. 12-11

- (iii) When the circle is tilted, so as to make 30° angle with the V.P., its top view will become inclined at 30° to xy . In the front view all the points will move along their respective paths (parallel to xy). Reproduce the top view keeping the centre o at the same distance, viz. 20 mm from xy and inclined at 30° to xy [fig. 12-11(ii)].
- (iv) For the final front view, project all the points upwards from this top view and horizontally from the first front view. Draw a freehand curve through the twelve points $1'_1, 2'_1$ etc. This curve will be an ellipse.

OBLIQUE PLANES PRACTICE DRAWING -3

Problem 12-6. (fig. 12-12): A square $ABCD$ of 50 mm side has its corner A in the H.P., its diagonal AC inclined at 30° to the H.P. and the diagonal BD inclined at 45° to the V.P. and parallel to the H.P. Draw its projections.

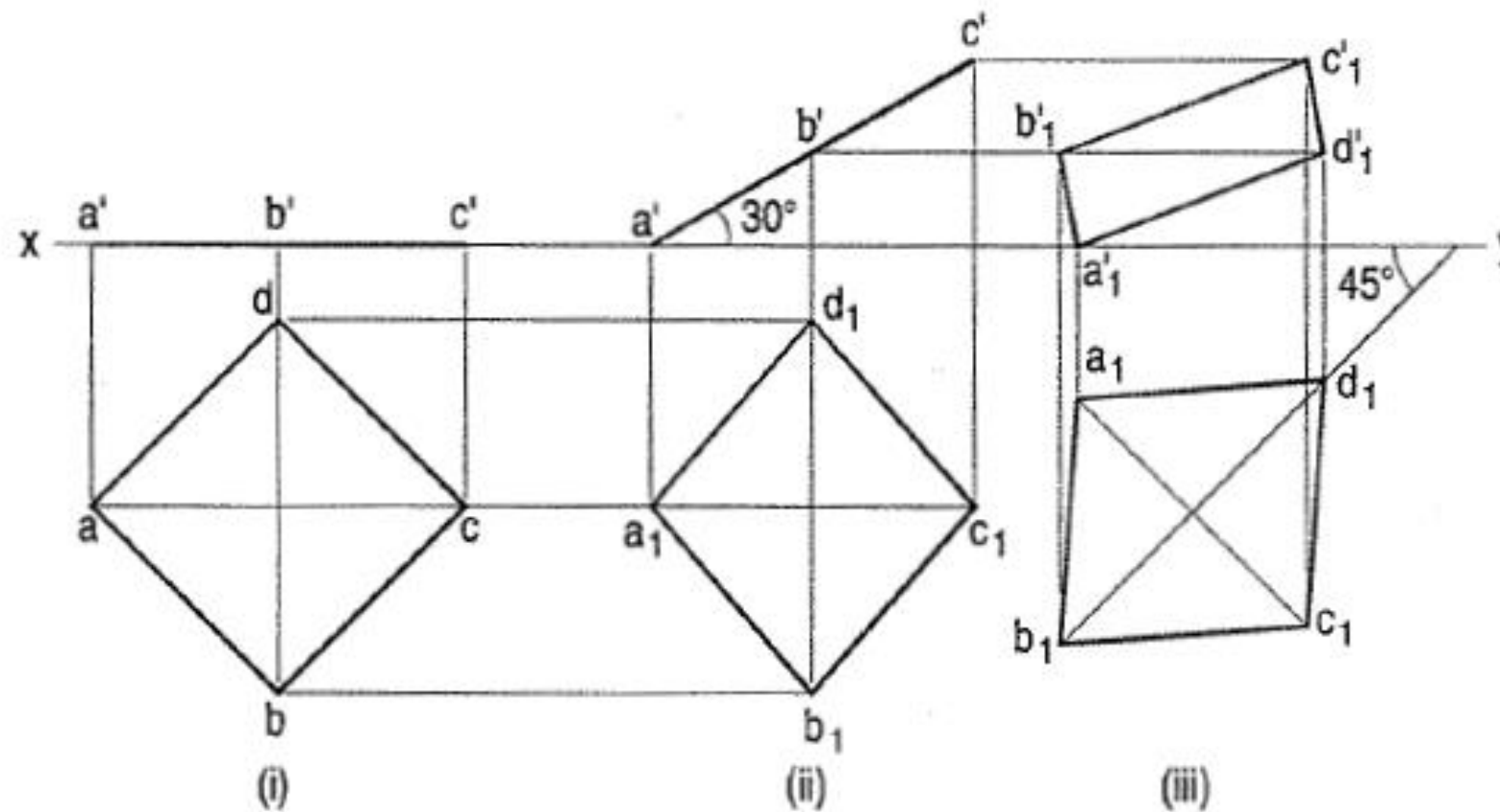


FIG. 12-12

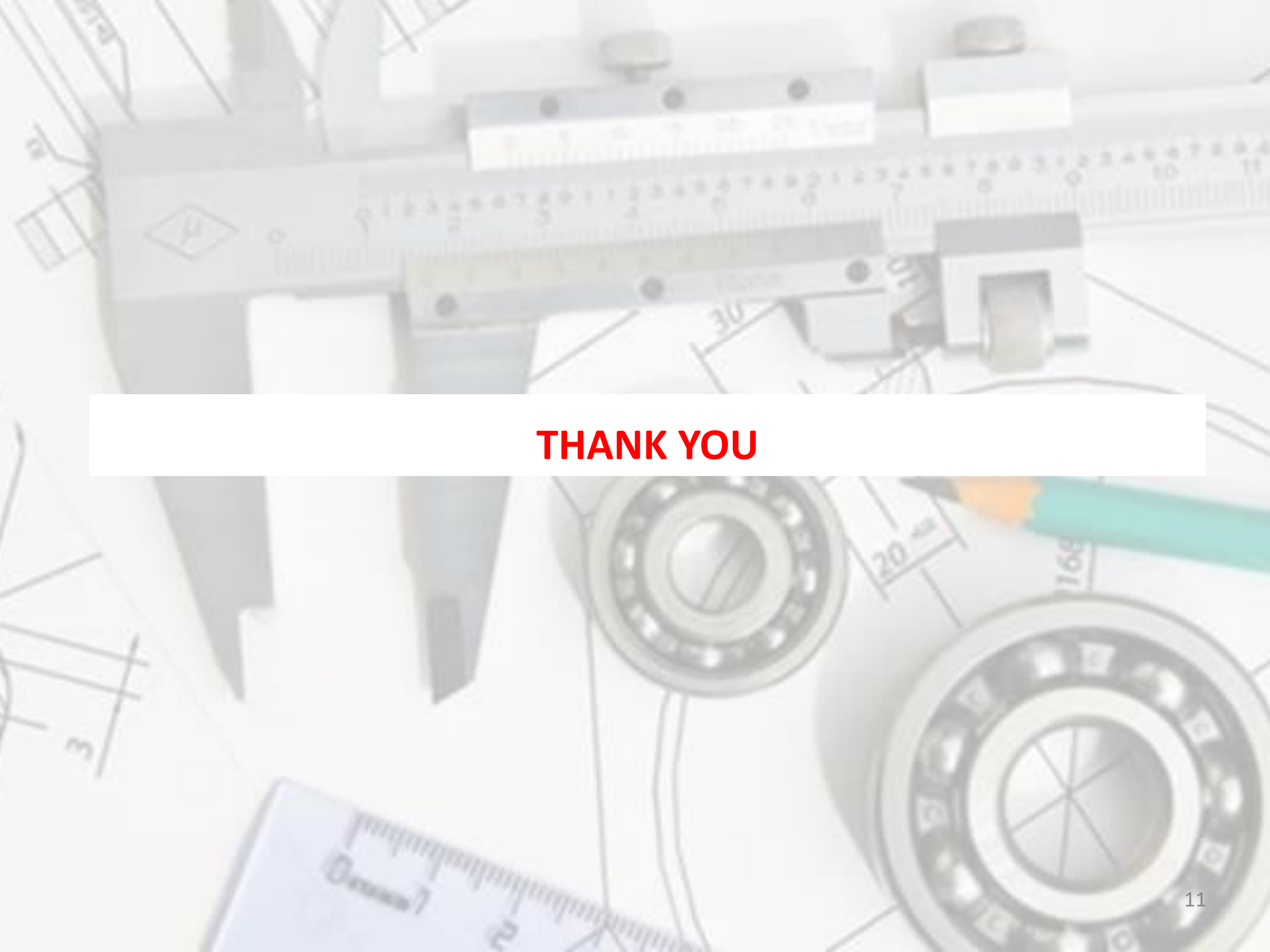
OBLIQUE PLANES

PRACTICE DRAWING -3

In the initial stage, assume the square to be lying in the H.P. with AC parallel to the V.P.

- (i) Draw the top view and the front view. When the square is tilted about the corner A so that AC makes 30° angle with the H.P., BD remains perpendicular to the V.P. and parallel to the H.P.
- (ii) Draw the second front view with $a'c'$ inclined at 30° to xy , keeping a' or c' in xy . Project the second top view. The square may now be turned so that BD makes 45° angle with the V.P. and remains parallel to the H.P. Only the position of the top view will change. Its shape and size will remain the same.

- (iii) Reproduce the top view so that b_1d_1 is inclined at 45° to xy . Project the final front view upwards from this top view and horizontally from the second front view.

The background is a collage of various mechanical and technical elements. At the top, there's a close-up of a metal component with a ruler and a small white label. Below that, a technical drawing of a bearing is visible. In the bottom right, there's a large, detailed drawing of a bearing with a green pencil pointing to it. A ruler is also visible in the bottom left corner. The overall theme is engineering and manufacturing.

THANK YOU